

## Logic PhD Exam, September 2000

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*Answer 7 questions, including at least one from each of the four segments.*

### 1. General Logic

1. State Gödel's incompleteness theorem, and briefly sketch its proof.
2. Show that any complete, finitely axiomatizable first-order theory is decidable.
3. Give an example of two countable models which are elementarily equivalent, but not isomorphic.

### 2. Model Theory

1. State and sketch a proof of the completeness theorem.
2. State and sketch a proof of the Łoś theorem about ultraproducts of models.

### 3. Set Theory

1. Prove that a countable union of countable sets is countable.
2. Sketch the proof that  $L \models \text{GCH}$ .
3. Prove that if  $\kappa$  is a regular, uncountable cardinal then any diagonal intersection of closed and unbounded subsets of  $\kappa$  is closed and unbounded.

### 4. Recursion Theory

1. Show that any total function  $f : \omega \rightarrow \omega$  with a recursively enumerable graph is recursive. Show that this is not true for partial functions.
2. Explain why, for any  $A \subset \omega$ , there are at most countably many sets  $B$  such that  $B <_T A$ .
3. State and prove Kleene's Recursion (or Fixed Point) Theorem.